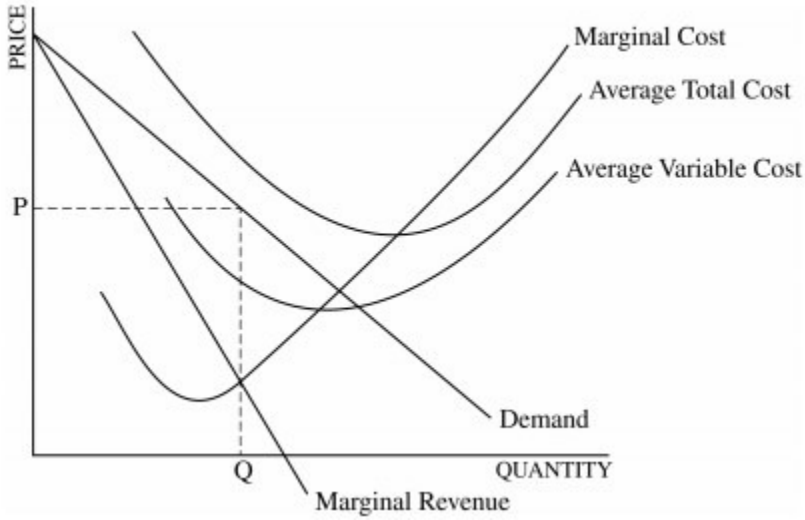
4.2 monopoly

ABBDB
DACBB
BDADC
ADCCD
BBECA
ADCCC

1.



The profit-maximizing firm depicted in the graph above should

(A) exit if conditions do not improve in the long run ✓

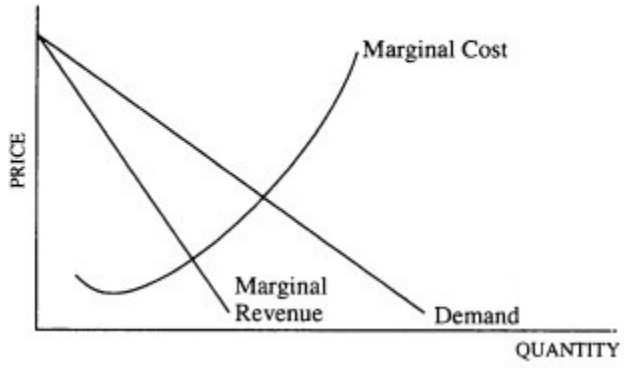
(B) produce the output that minimizes average total cost

(C) increase price to maximize profits

(D) increase output to maximize profits

(E) use less capital and more labor to reduce cost

2.



The graph above shows a firm's cost and revenue curves. This profit-maximizing firm will

(A) produce where demand is inelastic

(B) charge a higher price than that necessary to maximize revenues ✓

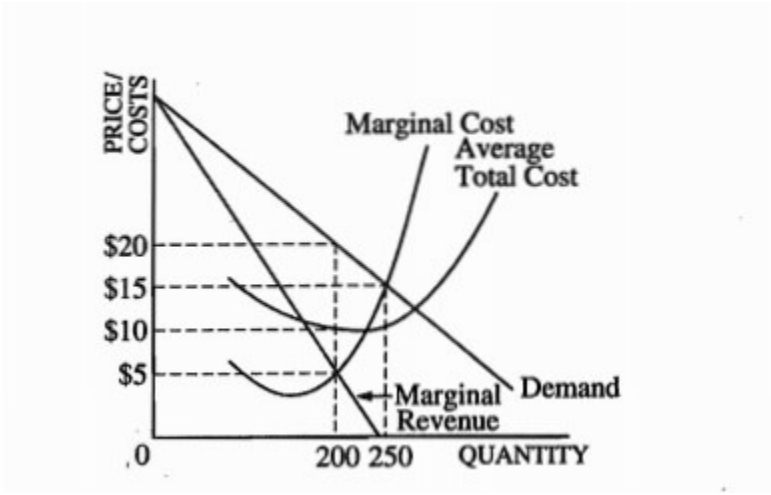
(C) have many profit-maximizing price and quantity combinations

(D) be unable to increase sales and total revenues by lowering its price

(E) never have a region of falling average total cost

4.2 monopoly

3.



Which of the following statements about the firm whose cost and revenue curves are shown above is correct?

(A) Its profit-maximizing price is $5.

(B) Its profit-maximizing output level is 200 units. ✓

(C) Its maximum profit is $4,000.

(D) If it produces 250 units, it will earn no economic profits.

(E) At the profit-maximizing level of output, its total cost is $1,000.

4. A monopolist introduces a technological innovation that lowers the marginal cost and average cost of production. The price of the good and the level of output are most likely to change in which of the following ways?

4.2 monopoly

(A)

Price	Level of Output
Remain constant	Remain constant

(B)

Price	Level of Output
Remain constant	Increase

(C)

Price	Level of Output
Increase	Decrease

(D) ✓

Price	Level of Output
Decrease	Increase

(E)

Price	Level of Output
Decrease	Remain constant

5. A monopolist is inefficient from society's point of view because

(A) it overproduces output and charges a price above marginal cost

(B) it underproduces output and charges a price above marginal cost ✓

(C) it sometimes colludes with other firms to obtain higher prices

(D) the price the monopolist charges is too low and it does not advertise

(E) consumers must pay whatever price the monopolist charges

6. A monopolist produces two unrelated goods, X and Y. The demand for X is currently price elastic and the demand for Y is currently price inelastic. To increase its total revenue, the firm should change the price of X and Y in which of the following ways?

4.2 monopoly

(A)

Price of X	Price of Y
Increase	No change

(B)

Price of X	Price of Y
Increase	Decrease

(C)

Price of X	Price of Y
Increase	Increase

(D) ✓

Price of X	Price of Y
Decrease	Increase

(E)

Price of X	Price of Y
Decrease	Decrease

7. A monopolist's demand curve is necessarily

(A) the same as the market demand curve ✓
(B) the same as its marginal revenue curve
(C) upward sloping
(D) perfectly elastic
(E) perfectly inelastic

8. A monopoly is different from a perfectly competitive firm in that a monopoly

4.2 monopoly

(A) does not have a U-shaped average total cost curve
(B) has an average fixed cost curve that is perfectly horizontal
(C) has a marginal revenue curve that lies below its demand curve ✓
(D) always earns economic profits
(E) operates in the inelastic segment of its demand curve

9. A profit-maximizing monopolist selects its output level in the

(A) inelastic region of its demand curve
(B) elastic region of its demand curve ✓
(C) range of output where marginal revenue is rising
(D) range of output where total cost is falling
(E) range of output where marginal cost is falling

10. A single-price monopolist is currently producing in the inelastic portion of its market demand curve. In order to maximize profits, the monopolist should change the price and output in which of the following ways?

4.2 monopoly

(A)

Price	Output
Increase	Increase

(B) ✓

Price	Output
Increase	Descrease

(C)

Price	Output
Increase	Not change

(D)

Price	Output
Descrease	Increase

(E)

Price	Output
Not change	Increase

11. A single-price monopolist's marginal revenue is

(A) equal to its price

(B) less than its price ✓

(C) greater than its price

(D) negative when it maximizes revenues

(E) zero when it maximizes profit

12. Assume that a monopolist is producing in the inelastic portion of its demand curve. Which of the following will occur if the monopolist decreases its price?

4.2 monopoly

(A) Marginal revenue will decrease, but profits will increase.
(B) Marginal revenue will increase, but profits will decrease.
(C) Total revenue will decrease, but profits will increase.
(D) Both total revenue and profits will decrease. ✓
(E) Both total revenue and profits will increase.

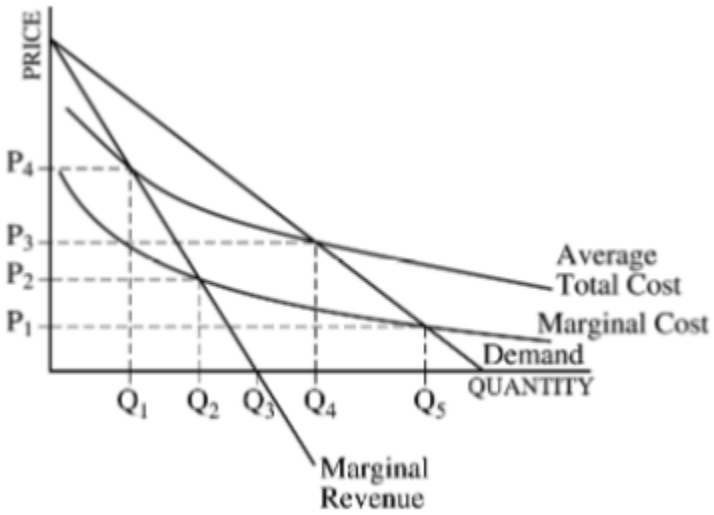
13. Assume that a profit-maximizing monopoly is charging a single price. If the monopoly can price discriminate and charge each consumer what he or she is willing to pay, which of the following will occur?

(A) The quantity of output produced will increase. ✓
(B) Total cost will decrease.
(C) Economic profit will decrease.
(D) Consumer surplus will increase.
(E) Demand will decrease.

14. Assuming a linear downward-sloping demand curve, as a monopoly firm sells additional units of output, its marginal revenue will

(A) increase at an increasing rate
(B) increase at first, then decrease
(C) decrease at first, then increase
(D) decrease continuously ✓
(E) remain constant

The question refers to the following diagram of a natural monopolist.



4.2 monopoly

15. The firm shown in the diagram qualifies as a natural monopoly because

(A) the demand curve is downward sloping

(B) the demand curve lies above the marginal revenue curve

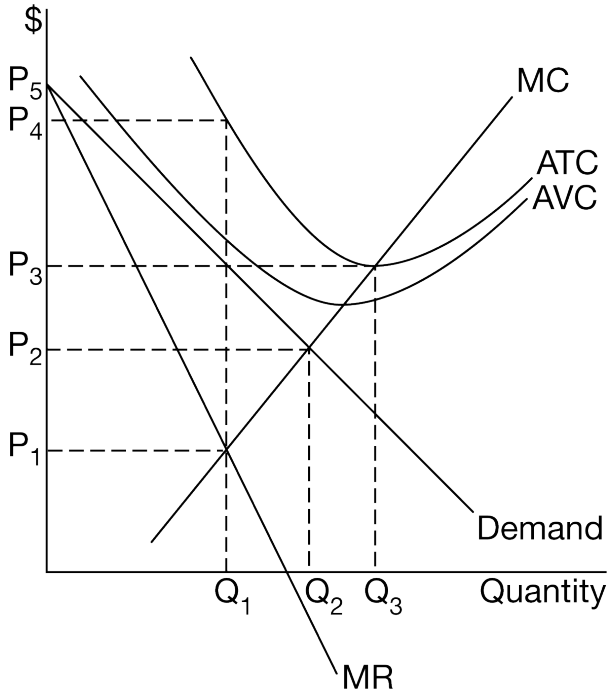
(C) the average total cost is decreasing in the relevant range of market demand ✓

(D) the firm can maximize profit with any output level it chooses

(E) marginal revenue is positive at the profit-maximizing output level

---

The following two questions refer to the cost and revenue conditions of a monopolistically competitive firm shown in the graph below.



MC = marginal cost, ATC = average total cost, AVC =average variable cost, and MR = marginal revenue.

16. The firm's profit-maximizing output in the short run is

(A) zero, because $\mathrm{P} < \mathrm{AVC}$ ✓

(B) Q_1, because $\mathrm{MR} = \mathrm{MC}$

(C) Q_2, because $\mathrm{P} = \mathrm{MC}$

(D) Q_3, because $\mathrm{MC} = \mathrm{ATC}$

(E) impossible to determine

4.2 monopoly

Answer A

Correct. The monopolist's profit-maximizing output is determined by equating marginal revenue to marginal cost. In the graph provided, marginal revenue equals marginal cost at Q_1 and the price is on the demand curve at P_3. However, at Q_1 the profit-maximizing price is less than both the ATC and the AVC. Therefore, the firm will minimize its losses by shutting down in the short run and exiting the market in the long run if conditions do not change. Thus, the profit-maximizing output in the short run is zero since the firm will shut down.

17. Which of the following will the firm do in the long run if market conditions do not change?

(A) It will increase output to Q_2 and lower price to P_2 to minimize losses.
(B) It will increase output to Q_3 and raise price to P_4 to earn zero economic profit.
(C) It will produce Q_1 and set price equal to marginal revenue.
(D) It will exit the industry. ✓
(E) It will build a larger plant to achieve decreasing returns to scale.

Answer D

Correct. In the graph provided, marginal revenue equals marginal cost at Q_1 and the price is on the demand curve at P_3. However, at Q_1 the profit-maximizing price is less than both the ATC and the AVC. Therefore, the firm will minimize its losses by shutting down in the short run and exiting the market in the long run if conditions do not change.

---

18. Which of the following is a source of monopoly power?

(A) Scarcity
(B) Elasticity of demand
(C) Barriers to entry ✓
(D) Low profits
(E) Free markets

19. Compared to a perfectly competitive industry with the same demand and cost curves, a monopoly's price and output will be which of the following?

4.2 monopoly

(A)

Price	Output
The same	Higher

(B)

Price	Output
Higher	The same

(C) ✓

Price	Output
Higher	Lower

(D)

Price	Output
Lower	The same

(E)

Price	**Output**
Lower	Higher

20. Compared with a perfectly competitive market,a single-price monopoly with the same market demand and cost curves will

(A) increase output and price

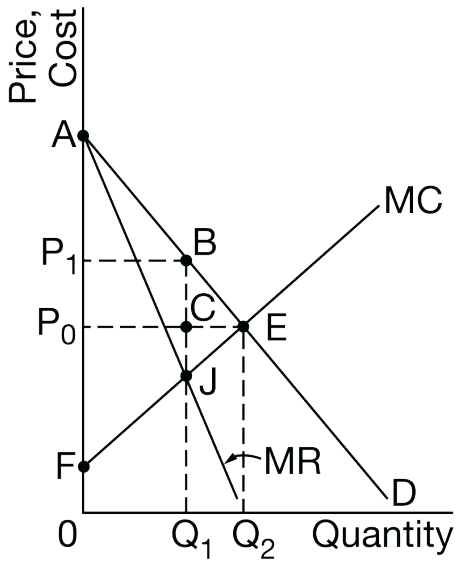
(B) increase output and decrease price

(C) decrease output and price

(D) decrease output and increase price ✓

(E) produce the same level of output and increase price

4.2 monopoly



Assume that the market is a profit-maximizing monopoly.

21. Which of the following areas shows the consumer surplus?

(A) $\mathrm{AP_0E}$

(B) $\mathrm{AP_1B}$ ✓

(C) AFE

(D) BJE

(E) $\mathrm{P_0FE}$

Answer B

Correct. The profit-maximizing monopolist would produce $\mathrm{Q_1}$ where $\mathrm{MR} = \mathrm{MC}$ and would set the price at $\mathrm{P_1}$. The consumer surplus is the area that is under the demand curve and above the monopoly price line at $\mathrm{Q_1}$, which is $\mathrm{AP_1B}$.

22. Which of the following areas shows the producer surplus and the deadweight loss?

4.2 monopoly

(A) The producer surplus is area P_1BCP_0, and the deadweight loss is area BJE.

(B) The producer surplus is area P_1BJF, and the deadweight loss is area BJE. ✓

(C) The producer surplus is area P_1BQ_10, and the deadweight loss is area BJE.

(D) The producer surplus is area P_1BQ_10, and the deadweight loss is area BCE.

(E) The producer surplus is area P_0EA, and the deadweight loss is area CEJ.

Answer B

Correct. With monopoly, the firm restricts output and raises price. The producer captures surplus from the consumer (P_0P_1BC), so its total producer surplus would be P_1BJF. Additionally, there is a deadweight loss equal to area BJE.

---

23. Which of the following can give a firm market power?

(A) Having access to common information

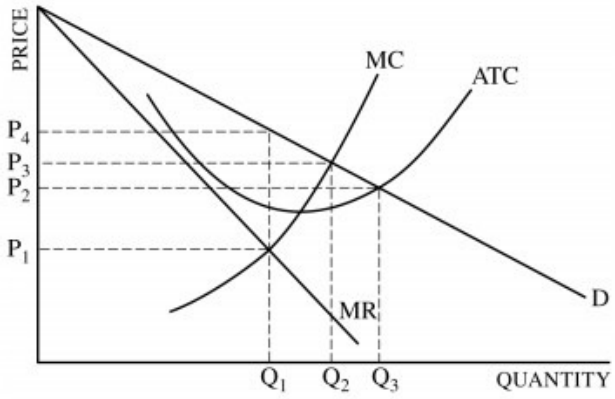
(B) Producing a standardized or homogeneous product

(C) Lacking barriers to entry or exit

(D) Having a large number of competitors in the market

(E) Having economies of scale in production over the range of market output ✓

The following questions refer to the monopoly graph below, where MC = marginal cost, ATC = average total cost, D = demand, and MR = marginal revenue.



24. The profit-maximizing combination of output and price for a single-price monopoly is

4.2 monopoly

(A) Q_1 and P_1
(B) Q_1 and P_2
(C) Q_1 and P_4 ✓
(D) Q_2 and P_3
(E) Q_3 and P_2

---

25. For an unregulated monopolist, the profit- maximizing quantity will always be

(A) in the elastic region of the demand curve ✓
(B) where marginal revenue equals price
(C) where price equals average total cost
(D) where price equals marginal cost
(E) where the marginal cost curve intersects the demand curve

26. If SteveR Incorporated is a monopolistic producer of diamonds, the firm's demand curve is down- ward sloping because

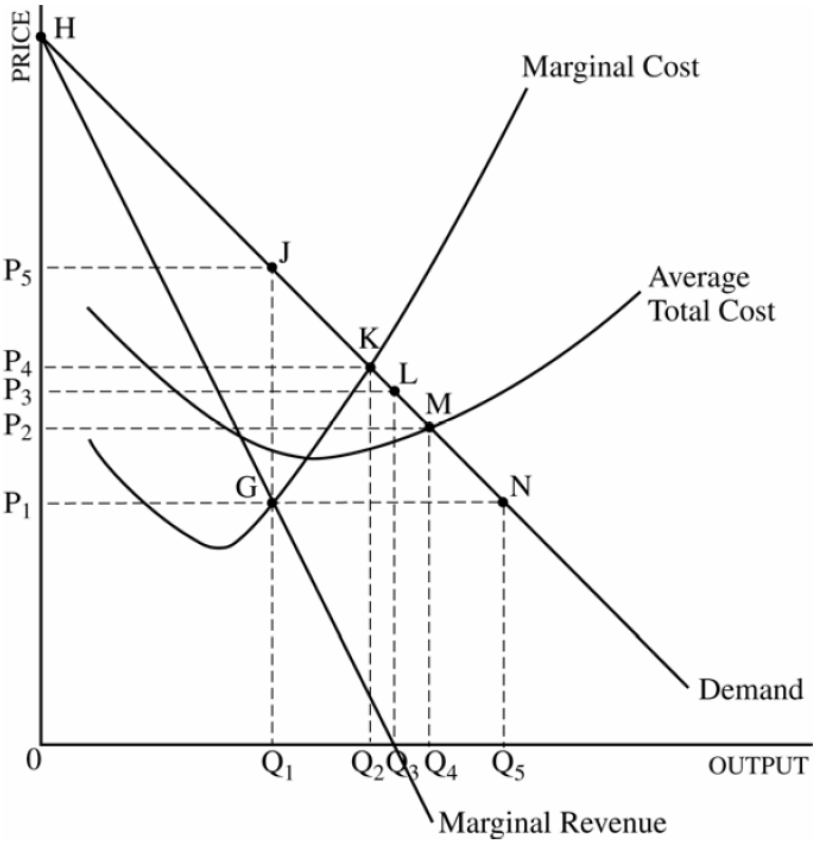
(A) the number of diamonds SteveR Incorporated offers for sale affects the price of diamonds ✓
(B) marginal revenue is negative throughout the range of the demand curve
(C) marginal revenue is positive throughout the range of the demand curve
(D) the diamond industry consists of a few firms selling similar diamonds
(E) the demand for diamonds is inelastic

27. From the point of view of economic efficiency, a monopolist produces

(A) too much of a good and charges too low a price
(B) too much of a good and charges too high a price
(C) too little of a good and charges too low a price
(D) too little of a good and charges too high a price ✓
(E) the socially optimal amount of a good

4.2 monopoly

The following questions refer to the diagram below, which shows the cost and revenue conditions of a monopolist.



28. If the monopolist chooses to maximize total revenue rather than total profit, it will choose which combination of price and output?

4.2 monopoly

(A)

Price	Output
P_1	Q_5

(B)

Price	Output
P_2	Q_4

(C) ✓

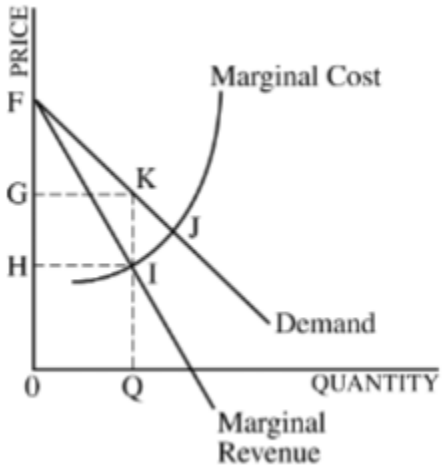
Price	Output
P_3	Q_3

(D)

Price	Output
P_4	Q_4

(E)

Price	Output
P_5	Q_5

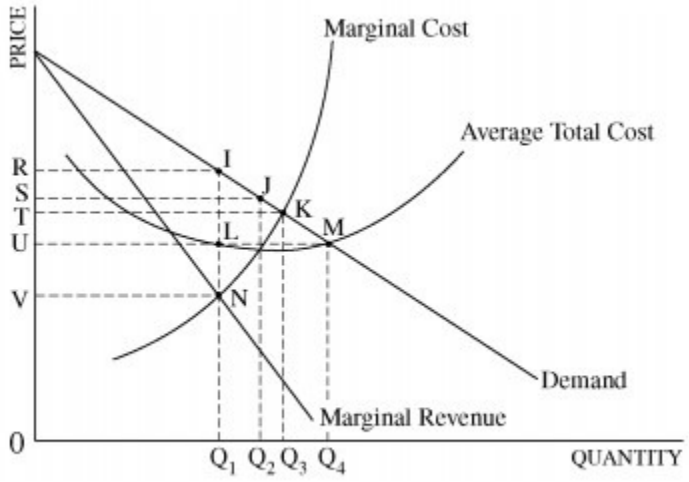


4.2 monopoly

29. In the diagram, the deadweight loss from a profit-maximizing monopolist is represented by area

(A) FGK
(B) FHI
(C) IJK ✓
(D) GHIK
(E) 0HIQ

The following questions are based on the graph below, which shows the cost and revenue curves of a monopoly firm.



30. The economic profit of the profit-maximizing monopolist is given by the area

(A) RSJI
(B) $R0Q_1I$
(C) RULI ✓
(D) RVNI
(E) $U0Q_4M$

31. If this were a perfectly competitive industry with the same costs as shown on the graph, the equilibrium price and output would be which of the following?

4.2 monopoly

(A)

Price	Output
0R	Q_1

(B)

Price	Output
0S	Q_2

(C) ✓

Price	Output
0T	Q_3

(D)

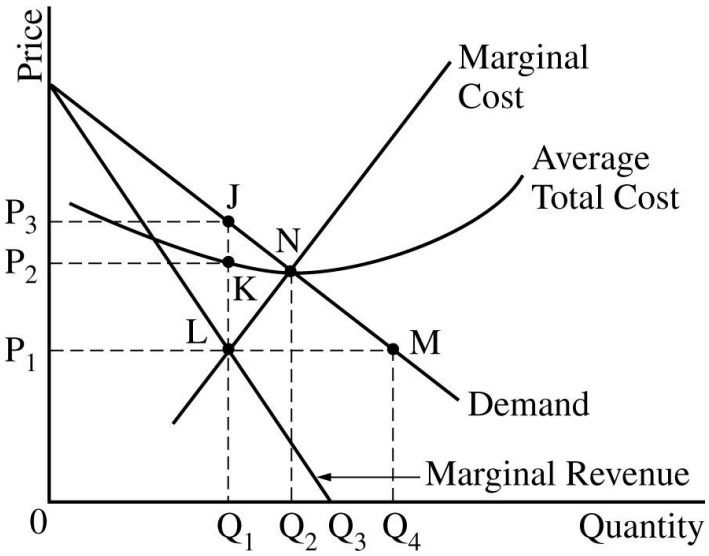
Price	Output
0U	Q_1

(E)

Price	Output
0U	Q_4

4.2 monopoly

The following questions refer to the graph of a profit- maximizing firm below.



32. The firm's economic profit is equal to

(A) area $0P_3\ JQ_1$

(B) area $0P_2\ KQ_1$

(C) area $P_2\ P_3\ JK$ ✓

(D) area $P_1\ P_3\ JL$

(E) zero

33. The firm's total revenue is equal to

(A) area P_1P_2KL

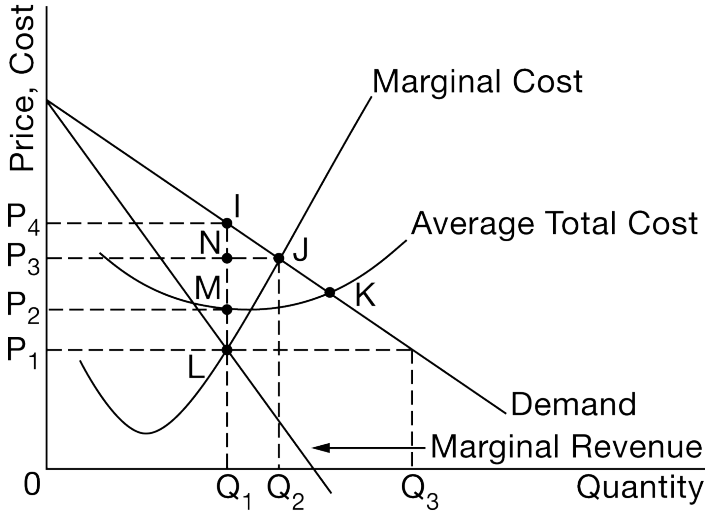
(B) area P_1P_3JL

(C) area $0P_3JQ_1$ ✓

(D) area $0P_2KQ_1$

(E) area JLM

4.2 monopoly



34. Which of the following combinations of output, price, and economic profit is consistent for the profit-maximizing monopolist depicted in the graph?

4.2 monopoly

(A)

Output	Price	Economic Profit
Q_1	P_1	$0P_1LQ_1$

(B)

Output	Price	Economic Profit
Q_1	P_4	P_1P_4IL

(C) ✓

Output	Price	Economic Profit
Q_1	P_4	P_2P_4IM

(D)

Output	Price	Economic Profit
Q_2	P_3	P_2P_3NM

(E)

Output	Price	Economic Profit
Q_3	P_1	P_1P_2ML

Answer C

Correct. The profit-maximizing output Q_1 occurs where $MC = MR$, and the price P_4 is above Q_1 on the demand curve. Economic profit is given by the area P_2P_4IM.

---

35. If a perfectly competitive industry were monopolized without any changes in cost conditions, the price and quantity produced would change in which of the following ways?

4.2 monopoly

(A)

Price	Quantity
Increase	Increase

(B) ✓

Price	Quantity
Increase	Decrease

(C)

Price	Quantity
Increase	May increase or decrease

(D)

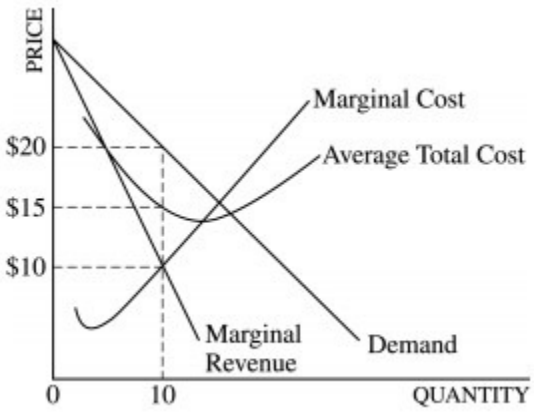
Price	Quantity
Decrease	Increase

(E)

Price	Quantity
Decrease	Decrease

4.2 monopoly

The following questions refer to the graph below.



36. If the firm produces 10 units of output, its economic profits will equal

(A) 0

(B) $50 ✓

(C) $100

(D) $150

(E) $200

37. If the marginal cost curve of a monopolist shifts up, which of the following will occur to the monopolist's price and output?

4.2 monopoly

(A)

Price	Output
Decrease	Increase

(B)

Price	Output
Decrease	Decrease

(C)

Price	Output
Increase	No change

(D)

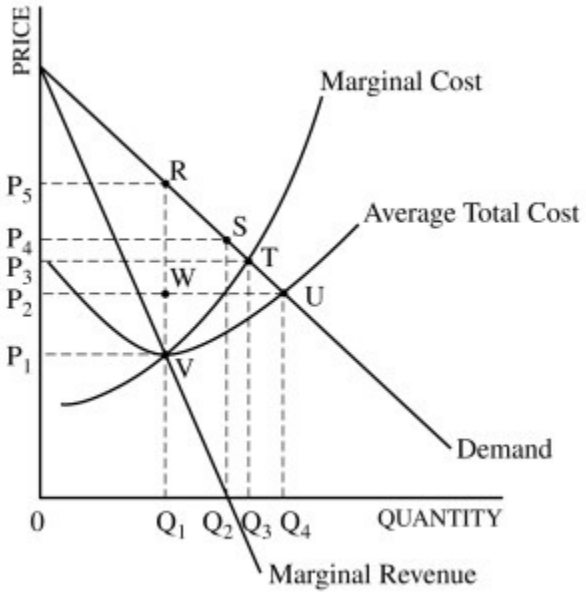
Price	Output
Increase	Increase

(E) ✓

Price	Output
Increase	Decrease

4.2 monopoly

The following questions refer to the graph of a monopolist shown below



38. If the monopolist is unregulated, its profitmaximizing price and output level would lead to a deadweight loss equal to area

(A) RUV

(B) RTV ✓

(C) RTW

(D) TUV

(E) UVW

39. Which of the following statements about cost is always true for both monopolies and perfectly competitive firms?

(A) Average total cost equals marginal cost when average total cost is a minimum. ✓

(B) Marginal cost decreases as production increases.

(C) Average fixed cost is equal to marginal cost when average fixed cost is a minimum.

(D) Average variable cost is equal to marginal cost when marginal cost is a minimum.

(E) Average variable cost decreases as production increases.

40. At the current quantity that a firm is selling, the firm has marginal revenue of $750 and marginal cost of $800. Which of the following is true?

4.2 monopoly

(A) The firm is maximizing profit.
(B) The firm's profits would increase if the firm increased the quantity sold.
(C) The firm's profits would increase if the firm decreased the quantity sold. ✓
(D) The firm earns negative economic profit.
(E) The firm earns zero accounting profit.

Answer C

Correct. The firm is producing where marginal cost is greater than marginal revenue. Profit is maximized when marginal revenue equals marginal cost, so the firm will increase its profits by decreasing output to where marginal revenue equals marginal cost.

41. Monopolies are inefficient compared to perfectly competitive firms because monopolies
(A) produce output with average total cost exceeding average revenue
(B) produce more output than is social desirable
(C) charge a price less than marginal revenue
(D) charge a price greater than marginal cost ✓
(E) charge a price less than average total cost

42. As the quantity of a good increases, which of the following is true in the elastic region of a monopolist's demand curve?
(A) Marginal revenue and total revenue are negative.
(B) Marginal revenue is increasing, and total revenue is positive.
(C) Marginal revenue is decreasing, and total revenue is negative.
(D) Marginal revenue is positive, and total revenue is increasing. ✓
(E) Marginal revenue is negative, and average revenue is decreasing.

Answer D

Correct. In the elastic region of the monopolist's demand curve, marginal revenue is positive, and decreasing. Total revenue is always positive and increases as quantity increases, because the percentage change in the quantity is greater than the percentage change in the price along the elastic region of the demand curve.

4.2 monopoly

43. Which of the following is true if a monopolist's marginal revenue is negative at the current level of output?

(A) Demand for its product is unit elastic.
(B) Demand for its product is price elastic.
(C) Demand for its product is price inelastic. ✓
(D) Marginal cost is equal to price.
(E) Marginal revenue is equal to price.

44. Which of the following is necessarily true at a monopolist's profit-maximizing level of output?

(A) Marginal revenue is equal to marginal cost, but greater than price.
(B) Marginal revenue is equal to marginal cost, but less than price. ✓
(C) Marginal revenue is equal to both marginal cost and price.
(D) Average total cost is minimized.
(E) Average variable cost is minimized.

45. Which of the following is true in the inelastic region of a monopolist's demand curve as output increases?

(A) Marginal revenue and total revenue are negative.
(B) Marginal revenue is increasing and total revenue is positive.
(C) Marginal revenue is decreasing and total revenue is negative.
(D) Marginal revenue is positive and total revenue is increasing.
(E) Marginal revenue is negative and total revenue is decreasing. ✓

Answer E

Correct. Marginal revenue is zero when total revenue is maximized. This occurs at the point of unit elasticity on the demand curve. Moving down the demand curve from the point of unit elasticity, the region of inelastic demand begins, marginal revenue is negative, and total revenue decreases. The monopolist will avoid producing along the inelastic portion of the demand curve by decreasing output until marginal revenue is zero or positive.

46. A monopoly is producing the allocatively efficient output if, for the last unit produced,

4.2 monopoly

(A) price equals minimum average total cost

(B) price equals marginal cost ✓

(C) marginal cost equals marginal revenue

(D) marginal cost equals average variable cost

(E) marginal revenue equals average variable cost

Answer B

Correct. Allocative efficiency means producers supply the quantity of each product that consumers demand, this is achieved when the monopoly charges a price equal to marginal cost. A perfectly price discriminating monopoly produces to where the price (of the last unit) equals marginal cost. There is no allocative inefficiency with perfect price discrimination.

47. Which of the following is true when a profit-maximizing monopolist produces in the elastic portion of its demand curve?

(A) It can increase total revenue by raising price.

(B) It can decrease average total cost by reducing output.

(C) Price is equal to marginal revenue.

(D) Marginal revenue is less than marginal cost.

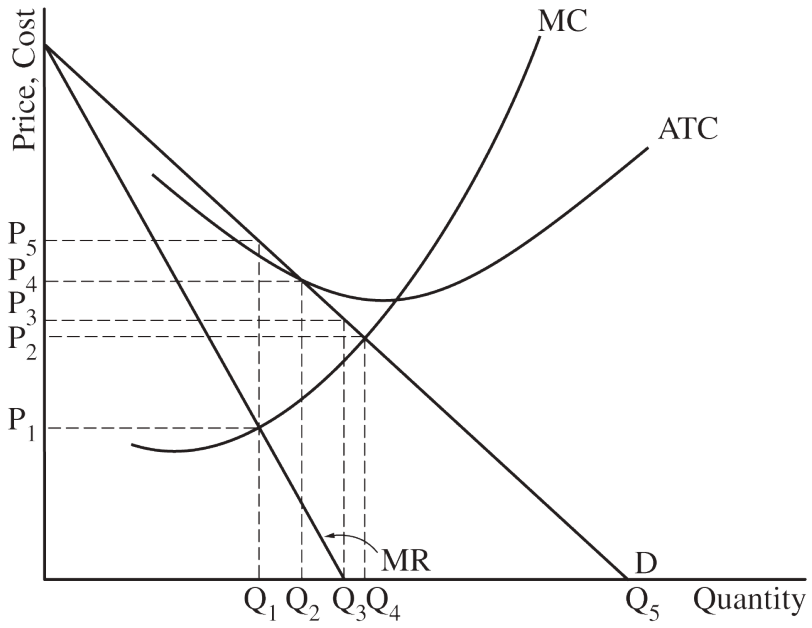
(E) Marginal revenue is positive. ✓

Answer E

Correct. Marginal revenue decreases with output but is positive in the elastic portion of the demand curve. That is, as price decreases, total revenue increases by smaller and smaller amounts and reaches a maximum when marginal revenue is zero and demand becomes unit elastic.

4.2 monopoly

Use the following graph to answer the questions below.



48. What price will the firm charge?

(A) P_1
(B) P_2
(C) P_3
(D) P_4
(E) P_5 ✓

Answer E

Correct. P_5 is determined on the demand curve above the quantity where marginal revenue equals marginal cost. The quantity (output) is determined by the intersection of marginal revenue and marginal cost.

49. What is the firm's profit-maximizing quantity of output?

4.2 monopoly

(A) Q_1 ✓

(B) Q_2

(C) Q_3

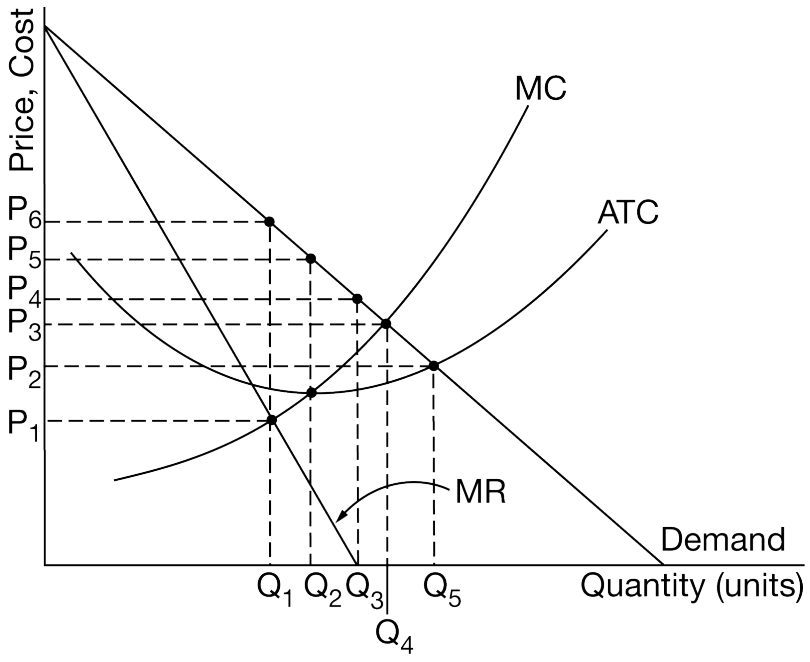
(D) Q_4

(E) Q_5

Answer A

Correct. Q_1 corresponds to the quantity at which marginal revenue equals marginal cost. This profit-maximizing rule, $MR=MC$, applies to all firms, whether perfect competition, monopoly, monopolistic competition, or oligopoly, as long as producing is preferable to shutting down. If price does not equal or exceed average variable cost, then the firm will shut down rather than produce the quantity at which $MR=MC$.

50.



What are the profit- and revenue-maximizing quantities for the monopolist whose revenue and cost conditions are described by the graph above?

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(A)

Profit-Maximizing	Revenue-Maximizing
Q_5	Q_2

(B)

Profit-Maximizing	Revenue-Maximizing
Q_4	Q_3

(C)

Profit-Maximizing	Revenue-Maximizing
Q_3	Q_3

(D)

Profit-Maximizing	Revenue-Maximizing
Q_1	Q_4

(E) ✓

Profit-Maximizing	Revenue-Maximizing
Q_1	Q_3

51. Which of the following is true for a monopoly but NOT for a perfectly competitive firm?

(A) The firm maximizes profit by equating marginal cost to marginal revenue.

(B) The firm's demand curve is the same as its average revenue curve.

(C) At the profit-maximizing output level, price is less than marginal revenue.

(D) The firm faces a downward-sloping demand curve. ✓

(E) The firm earns zero economic profit in the long run.

52. The profit-maximizing output level produced by an unregulated monopoly is

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(A) the socially optimal output level, since the firm's marginal revenue equals its marginal cost
(B) greater than the socially optimal level, since the firm's marginal cost exceeds its marginal revenue
(C) greater than the socially optimal level, since the firm makes economic profits
(D) less than the socially optimal level, since the price paid by consumers exceeds the firm's marginal cost ✓
(E) less than the socially optimal level, since the price of the product is less than the firm's marginal revenue

53. Which of the following is most likely to occur if a single-price monopolist is replaced by a perfectly competitive market?

(A) Prices will increase.
(B) The deadweight loss will decrease. ✓
(C) Profits will increase.
(D) Output will decrease.
(E) The firm’s cost curves will shift upward.

54. Most economists argue that a monopoly is inefficient because it

(A) has no incentive to minimize its costs
(B) produces too little output and sets a price above marginal cost ✓
(C) earns too much profit by charging consumers any price it wants
(D) produces too much output and thus wastes scarce resources
(E) usually produces unsafe products if not regulated by government

55. Which of the following is always true of a monopoly that is producing a level of output such that marginal revenue is negative?

(A) It is experiencing economies of scale in production.
(B) It is maximizing total revenue.
(C) It is producing where demand is price elastic.
(D) It could lower price to increase profits.
(E) It could decrease output to increase profits. ✓

56. Within the range of market demand, which of the following is consistent with the conditions of a natural monopoly?

(A) Long-run total cost decreases as output increases.
(B) Long-run average total cost remains constant as output increases.
(C) Long-run average total cost decreases as output increases. ✓
(D) Marginal cost exceeds average cost.
(E) Setting price equal to marginal cost will maximize profits.

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57. Regulating a natural monopoly to set its output price equal to the marginal cost of production will result in economic losses because for the last unit of output produced, marginal cost is

(A) less than the marginal willingness to pay and less than average total cost

(B) greater than the marginal willingness to pay and less than average total cost

(C) equal to the marginal willingness to pay and less than average total cost ✓

(D) equal to the marginal willingness to pay and equal to average total cost

(E) greater than the marginal willingness to pay and equal to average total cost

Answer C

Correct. Marginal willingness to pay is determined by the price along the demand curve. When marginal cost is equal to the price, then it is also equal to the marginal willingness to pay (demand). For a natural monopoly, average total cost decreases along the relevant range of demand, therefore marginal cost must lie below the average total cost along that range. Therefore, for the last unit of output produced, marginal cost will be equal to the marginal willingness to pay and less than the average total cost, since the firm is earning economic losses.

58. Which of the following statements concerning a natural monopoly is true?

(A) Average total cost is always less than marginal cost in the long run.

(B) At the allocatively efficient level of output, monopoly profit and deadweight loss are both equal to zero.

(C) If the monopolist chooses to produce the quantity at which price is equal to average cost, it would earn a normal profit. ✓

(D) The monopolist can earn positive economic profits by producing the allocatively efficient output in the short run.

(E) Production efficiency could be improved if another firm were to enter and compete with the single monopolist.

Answer C

Correct. With a natural monopoly, there are economies of scale (falling long-run average cost) over the range of the effective demand. Thus, a single firm can supply the entire market at a lower average total cost than two or more competing firms sharing the market. If the monopoly chooses to set the price equal to average total cost and avoid regulation, it would earn a normal or zero economic profit.

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59. Which of the following is true of a natural monopoly?

(A) The average total cost is constant throughout the entire effective demand.
(B) Marginal cost decreases throughout the entire effective demand.
(C) The average total cost initially decreases and then increases throughout the entire effective demand.
(D) The marginal cost initially increases and then decreases throughout the entire effective demand.
(E) The average total cost decreases throughout the entire effective demand. ✓

Answer E

Correct. For a natural monopoly, economies of scale exist throughout the effective demand so average total cost must decrease throughout the effective demand.

60. Which of the following must be true for a firm that is a natural monopoly?

(A) It can produce and supply its product to an entire market at a lower cost than could a number of smaller firms. ✓
(B) It produces at the minimum of its average total cost curve.
(C) It has a patent on its product.
(D) It will not be able to maximize profits unless subsidized by the government.
(E) It is productively efficient for the government to break up the monopoly into smaller firms.

61. A natural monopoly exists when a single firm

(A) produces a quantity of output that maximizes the total producer surplus
(B) produces the allocatively efficient quantity of output of a good
(C) has an upward-sloping marginal cost curve
(D) has a downward-sloping average total cost curve over the relevant range of output ✓
(E) sets price equal to marginal cost in order to maximize profit

Answer D

Correct. A natural monopoly is when the entire demand within a relevant market can be satisfied at the lowest cost by one firm rather than by two or more firms. As a result, the firm has a cost advantage and its average total cost decreases as it produces more output, leading to a downward-sloping average total cost curve throughout the relevant range of output.

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62. One difference between a firm in a perfectly competitive market and an unregulated monopoly is that the

(A) perfectly competitive firm can increase the quantity it sells at the market price, whereas the monopoly must lower its price to sell more ✓

(B) perfectly competitive firm sells differentiated products, whereas the monopoly sells a homogeneous product

(C) price elasticity of demand for a monopoly is much higher than that for a perfectly competitive firm

(D) monopolist seeks to maximize profits, whereas the perfectly competitive firm maximizes revenues

(E) monopolist can never change a price, whereas the perfect competitor can

63. Which of the following statements is true for a monopolist at the profit-maximizing output level?

(A) Price exceeds marginal revenue. ✓

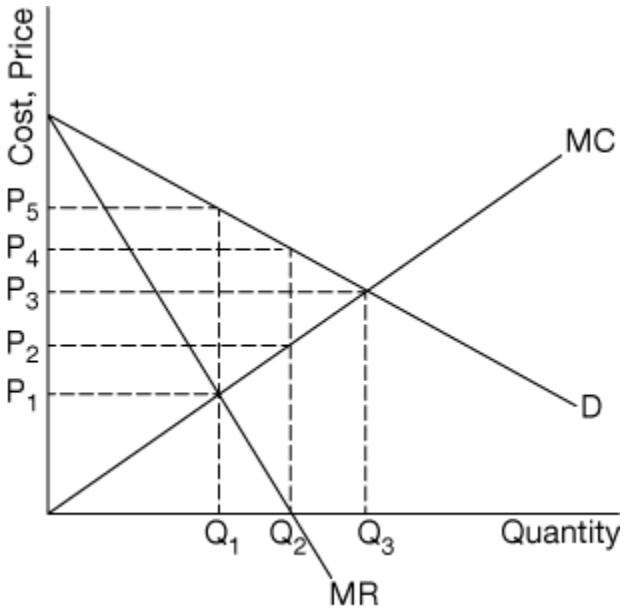
(B) Marginal costs exceeds price.

(C) Demand is price inelastic.

(D) Price equals marginal cost, which equals average total cost.

(E) The demand curve intersects the supply curve.

64.



Given the information in the graph provided, what is the monopolist's profit-maximizing quantity and price combination?

(A) Q_1, P_1

(B) Q_3, P_3

(C) Q_2, P_2

(D) Q_2, P_4

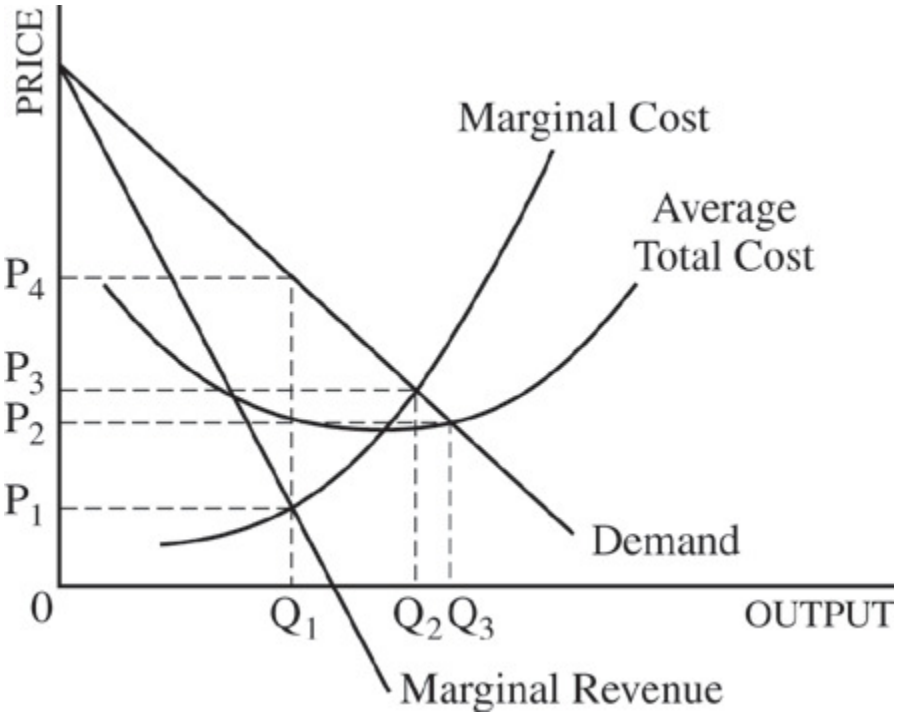
(E) Q_1, P_5 ✓

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Answer E

Correct. To maximize profits, firms should produce the quantity at which marginal revenue equals marginal cost $(\mathrm{MR} = \mathrm{MC})$. This is achieved when the firm produces Q_1 The monopolist sets the price from the demand curve at P_5.

65.



For the firm shown in the graph above, which combination of output and price will maximize its profit?

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(A)

Output	Price
Q_1	P_4

✓

(B)

Output	Price
Q_1	P_1

(C)

Output	Price
Q_2	P_3

(D)

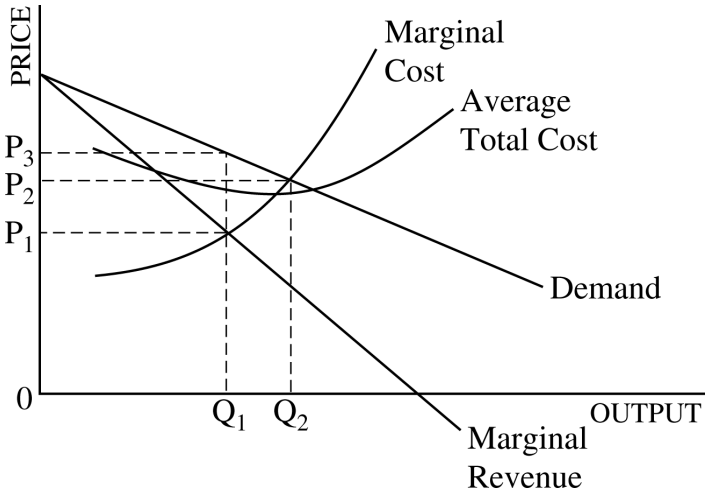
Output	Price
Q_2	P_2

(E)

Output	Price
Q_3	P_2

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66.



For the firm shown in the graph above, the short- run, profit-maximizing strategy would be to set output at

(A) Q_1, price at P_1, and suffer a loss

(B) Q_1, price at P_3, and earn an economic profit ✓

(C) Q_1, price at P_3, and earn only a normal profit

(D) Q_2, price at P_2, and earn an economic profit

(E) Q_2, price at P_2, and earn only a normal profit

67. Prosper Company is a profit-maximizing monopoly producing toy cubes. Prosper Company is currently earning positive economic profit. Which of the following combinations of events will definitely cause Prosper Company's profit to increase?

(A) A decrease in demand for toy cubes and a decrease in fixed costs to produce toy cubes

(B) A decrease in demand for toy cubes and an increase in fixed costs to produce toy cubes

(C) A decrease in demand for toy cubes and a decrease in variable costs to produce toy cubes

(D) An increase in demand for toy cubes and an increase in government subsidies to produce toy cubes ✓

(E) An increase in demand for toy cubes and an increase in fixed costs to produce toy cubes

Answer D

Correct. An increase in demand will increase both price and quantity of units sold (increasing total revenue). An increase in government subsidies will lower total costs for Prosper Company. This combination of an increase in total revenue and a decrease in total costs will cause Prosper Company's profits to increase.

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68. Which of the following is necessarily true of the profit-maximizing equilibrium of a monopolist who sets a single price?

(A) Price equals average total cost.

(B) Price is greater than marginal cost. ✓

(C) Average total cost is at its minimum level.

(D) Marginal revenue is greater than marginal cost.

(E) Marginal cost is minimized.

69. Which of the following statements is true for a perfectly competitive firm but **not** true for a monopoly?

(A) The firm's price is equal to its average revenue.

(B) The firm cannot affect the market price for its good. ✓

(C) It is difficult for other firms to enter the industry.

(D) The demand for the firm's product is unit elastic.

(E) The firm must lower its price in order to sell more of its product.